Social Network Analysis (CS342) | Assignment 1

Basic Concepts and Exploration of NetworkX in Python

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Task 1: Import networkx library (and other required libraries)

```
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np
print("Libraries imported.")
```

```
Libraries imported.
```

Task 2: Create an empty graph and add nodes

```
G = nx.Graph() #empty graph

G.add_node(1) # individual nodes
G.add_node(2)
G.add_nodes_from([3, 4, 5]) # adding additional nodes (list)

print("Nodes in Graph G:", G.nodes())
```

```
Nodes in Graph G: [1, 2, 3, 4, 5]
```

Task 3: Add edges to above graph

```
# Adding edges from a node to another node individually
G.add_edge(1, 2) # node 1 to 2
G.add_edge(1, 3) # node 1 to 3

G.add_edges_from([(2, 3), (2, 4), (3, 5), (4, 5)]) # adding multiple
edges using a list of tuples

print("Edges in Graph G:", G.edges())
```

```
Edges in Graph G: [(1, 2), (1, 3), (2, 3), (2, 4), (3, 5), (4, 5)]
```

Task 4: Create following types of graph:

- Undirected Graph
- Directed Graph
- Multigraph
- Directed Multigraph
- Weighted Grap

```
# 1. Undirected Graph (Already created as G)
print("Undirected Graph G created.")

# 2. Directed Graph
DG = nx.DiGraph()
DG.add_edges_from([(1, 2), (2, 1), (2, 3), (3, 4)])
print("Directed Graph DG created.")

# 3. Multigraph (Allows multiple edges between same pair of nodes)
MG = nx.MultiGraph()
MG.add_edges_from([(1, 2), (1, 2), (2, 3)])
print("Multigraph MG created (note parallel edges between 1 and 2).")

# 4. Directed Multigraph
DMG = nx.MultiDiGraph()
DMG.add_edges_from([(1, 2), (1, 2), (2, 1)])
print("Directed Multigraph DMG created.")

# 5. Weighted Graph
WG = nx.Graph()
WG.add_edge(1, 2, weight=1.0)
WG.add_edge(2, 3, weight=1.2)
WG.add_edge(3, 4, weight=1.5)
print("Weighted Graph WG created.")
```

```
Undirected Graph G created.
Directed Graph DG created.
Multigraph MG created (note parallel edges between 1 and 2).
Directed Multigraph DMG created.
Weighted Graph WG created.
```

Task 5 & 6: Remove edges and Display Adjacency Matrices

Removing edges and nodes, and showing the adjacency matrix before and after modification.

```
def display_adj_matrix(graph, name):
    print(f"\n Adjacency Matrix for {name} ")
    try:
        # Use simple dense conversion for display
        matrix = nx.adjacency_matrix(graph).todense()
        print(matrix)
```

```python
    except Exception as e:
        print(f"Could not display matrix: {e}")

# Display Initial Matrices
display_adj_matrix(G, "Undirected Graph G (Initial)")
display_adj_matrix(DG, "Directed Graph DG (Initial)")

#  Task 5: Remove few edges
print("\n[Action] Removing edge (1, 2) from G")
G.remove_edge(1, 2)

print("[Action] Removing edge (1, 2) from DG")
DG.remove_edge(1, 2)

#  Task 6 Logic: Remove a node to delete it clearly
print("[Action] Removing node 3 from G")
G.remove_node(3)

# Display Matrices After Modifications
display_adj_matrix(G, "Undirected Graph G (After removing edge (1,2)
and node 3)")
display_adj_matrix(DG, "Directed Graph DG (After removing edge
(1,2))")
```

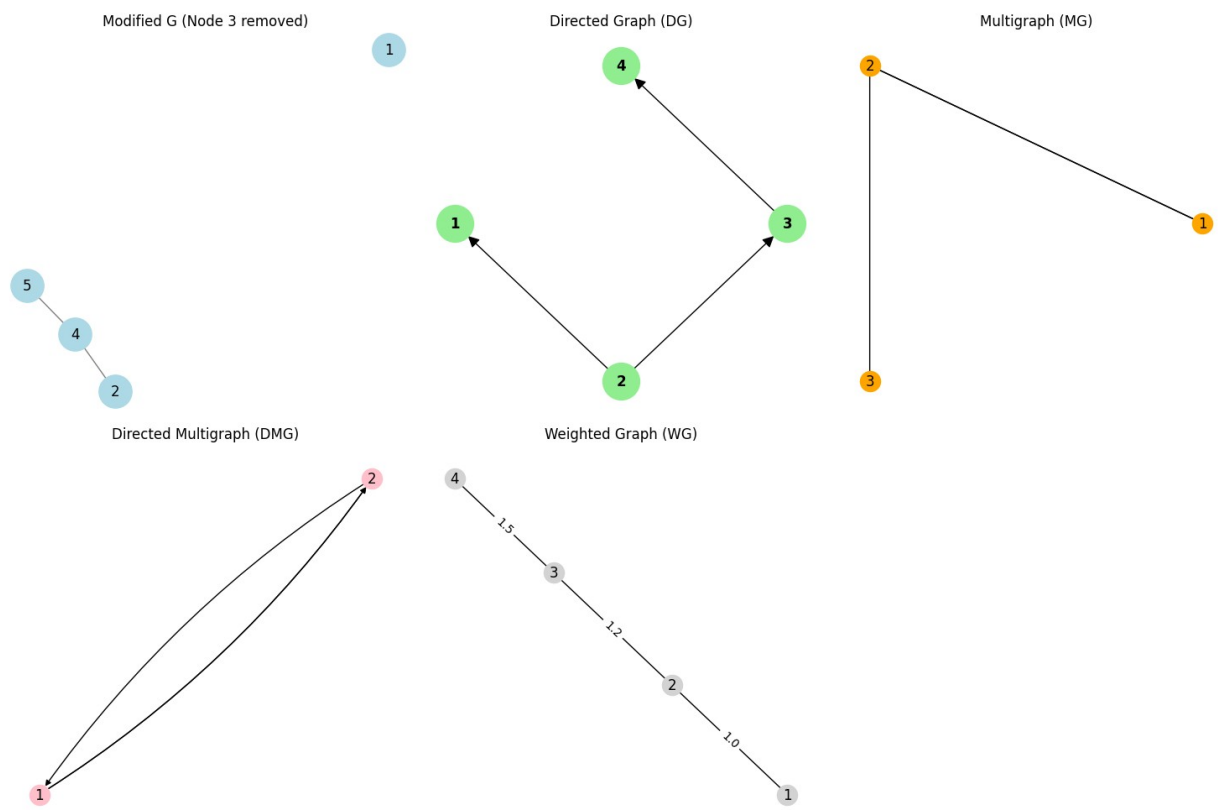
```
 Adjacency Matrix for Undirected Graph G (Initial)
[[0 1 1 0 0]
 [1 0 1 1 0]
 [1 1 0 0 1]
 [0 1 0 0 1]
 [0 0 1 1 0]]

 Adjacency Matrix for Directed Graph DG (Initial)
[[0 1 0 0]
 [1 0 1 0]
 [0 0 0 1]
 [0 0 0 0]]

[Action] Removing edge (1, 2) from G
[Action] Removing edge (1, 2) from DG
[Action] Removing node 3 from G

 Adjacency Matrix for Undirected Graph G (After removing edge (1,2)
and node 3)
[[0 0 0 0]
 [0 0 1 0]
 [0 1 0 1]
 [0 0 1 0]]

 Adjacency Matrix for Directed Graph DG (After removing edge (1,2))
```

```
[[0 0 0 0]
 [1 0 1 0]
 [0 0 0 1]
 [0 0 0 0]]
```

Task 7: Visualize graphs

Visualizing the previously created graphs with `networkx.draw()` functions.

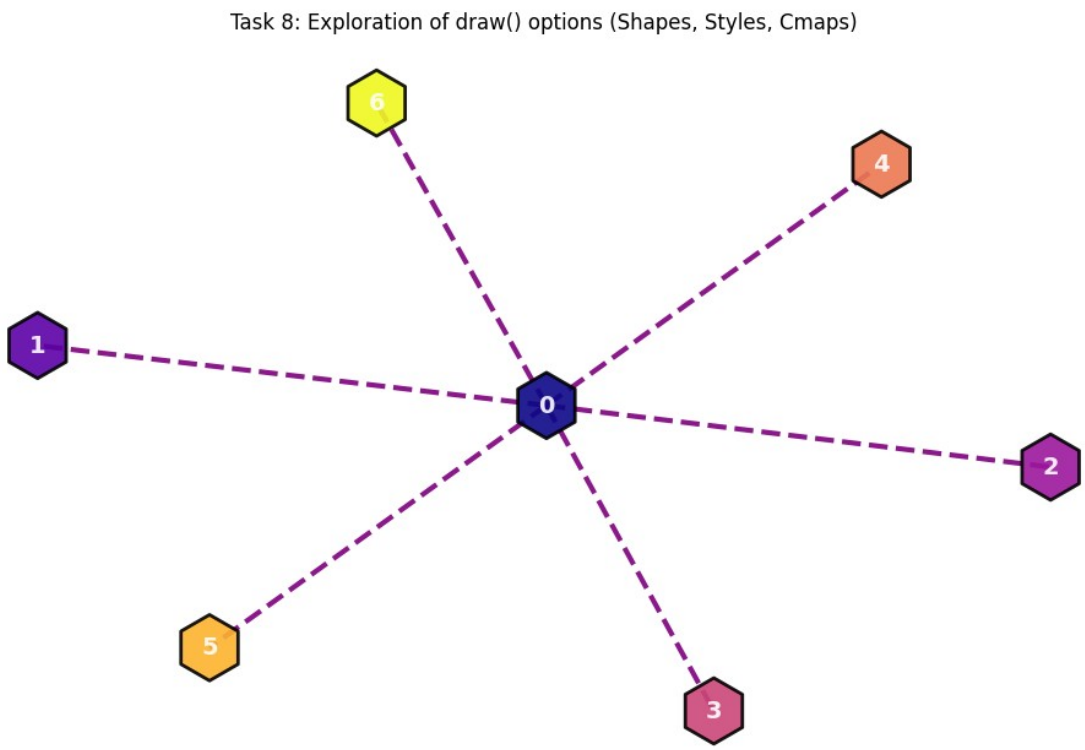
```
plt.figure(figsize=(15, 10))

# 1. Visualization of the modified Undirected Graph G
plt.subplot(2, 3, 1)
nx.draw(G, with_labels=True, node_color='lightblue',
edge_color='gray', node_size=800)
plt.title("Modified G (Node 3 removed)")

# 2. Directed Graph DG with custom styling options (Task 8
exploration)
plt.subplot(2, 3, 2)
# Usage of shell_layout, specific colors, and arrow styling
nx.draw(DG, pos=nx.shell_layout(DG), with_labels=True,
        node_color='lightgreen', font_weight='bold',
        node_size=1000, arrowsize=20, arrowstyle='-|>')
plt.title("Directed Graph (DG)")

# 3. Multigraph MG
plt.subplot(2, 3, 3)
# Circular layout
nx.draw_circular(MG, with_labels=True, node_color='orange')
plt.title("Multigraph (MG)")

# 4. Directed Multigraph DMG
plt.subplot(2, 3, 4)
nx.draw(DMG, with_labels=True, node_color='pink',
connectionstyle='arc3, rad = 0.1')
plt.title("Directed Multigraph (DMG)")

# 5. Weighted Graph WG
plt.subplot(2, 3, 5)
pos = nx.spring_layout(WG)
nx.draw(WG, pos, with_labels=True, node_color='#D3D3D3')
# Draw edge labels to show weights
edge_labels = nx.get_edge_attributes(WG, 'weight')
nx.draw_networkx_edge_labels(WG, pos, edge_labels=edge_labels)
plt.title("Weighted Graph (WG)")

plt.tight_layout()
plt.show()
```



Task 8: Explore different options in `networkx.draw()`

Exploring advanced options available in the drawing function, such as:

- **Node styling:** Shapes (`node_shape`), Colormaps (`cmap`), transparency (`alpha`), borders (`linewidths`, `edgecolors`).
- **Edge styling:** Styles (`style`), widths (`width`).
- **Font styling:** Size (`font_size`), color (`font_color`).

```
plt.figure(figsize=(10, 6))

# Create a graph for exploration (e.g., a Star Graph)
Explore_G = nx.star_graph(6)

# Exploring different options in nx.draw()
nx.draw(
    Explore_G,
    with_labels=True,

    # Node Styling
    node_size=1500,
    node_color=range(len(Explore_G)), # Numeric values mapping to cmap
    cmap=plt.cm.plasma,              # Color map
    node_shape='h',                  # Hexagon shape
    alpha=0.9,                       # Transparency
```

```
    linewidths=2,                       # Node border width
    edgecolors='black',                 # Node border color

    # Edge Styling
    width=3,                            # Edge width
    edge_color='purple',                # Edge color
    style='dashed',                     # Dashed line style

    # Font Styling
    font_size=14,
    font_color='white',
    font_weight='bold'
)

plt.title("Task 8: Exploration of draw() options (Shapes, Styles,
Cmaps)")
plt.show()
```